Closing the Computational-Statistical Gap in Best Arm Identification for Combinatorial Semi-bandits

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Introduction

1-page Summary

In combinatorial semi-bandits,...

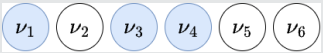
- most statistically efficient algorithms suffer computational inefficiency [Garivier and Kaufmann, 2016, Wang et al., 2021, Jourdan et al., 2021].
- most computationally efficient algorithms suffer statistical inefficiency [Katz-Samuels et al., 2020, Du et al., 2021].
- we fulfill the computational-statistical gap.

Combinatorial semi-bandits

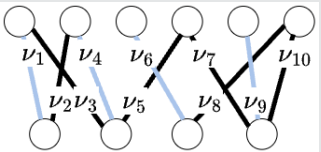
- K arms (K probability distributions), say $\nu_1 = \mathcal{N}(\mu_1, 1), \nu_2 = \mathcal{N}(\mu_2, 1) \ldots, \nu_K = \mathcal{N}(\mu_K, 1)$.
- Set of actions, $\mathcal{X} \subseteq \{0,1\}^K$, with a combinatorial structure.

Examples for combinatorial structures

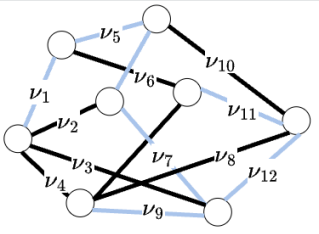
m-sets



matchings



spanning trees



In round t, an agent

1. pulls an action $\boldsymbol{x}(t) \in \mathcal{X}$
2. receives the reward $y_k(t) \sim \mathcal{N}(\mu_k, 1)$ if $x_k(t) = 1$

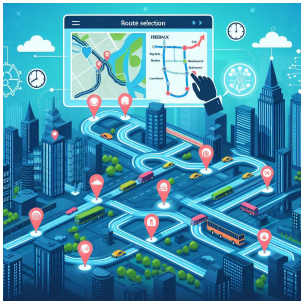
Best Action Identification with Fixed Confidence

Goal: identify the best action $i^\star(\boldsymbol{\mu}) \in \arg\max_{x \in \mathcal{X}} \langle \boldsymbol{x}, \boldsymbol{\mu} \rangle$.

A strategy consist of

- (sampling rule) $\boldsymbol{x}(t) \in \mathcal{F}_t := \sigma(\boldsymbol{x}(1), \boldsymbol{y}(1), \ldots, \boldsymbol{x}(t-1), \boldsymbol{y}(t-1))$ (arm to explore)
- (stopping rule) τ (round to stop)
- (decision rule) $\mathcal{F}_\tau$-measurable $\hat{\boldsymbol{\imath}} \in \mathcal{X}$ (guess of best action to return)

Wish to minimize $\mathbb{E}_{\boldsymbol{\mu}}[\tau]$ subject to $\mathbb{P}_{\boldsymbol{\mu}}[\hat{\boldsymbol{\imath}} \neq \boldsymbol{\imath}^\star(\boldsymbol{\mu})] \leq \delta$.



In traffic management, authorities might select a set of routes for traffic flow optimization. Feedback on each route (e.g., congestion levels, travel times) helps improve future route selections.

Information-theoretic Lower Bound [Garivier and Kaufmann, 2016]

For any δ-PAC algorithm (stop in finite time a.e. and $\mathbb{P}_{\boldsymbol{\mu}}[\hat{\boldsymbol{\imath}} \neq \boldsymbol{\imath}^\star(\boldsymbol{\mu})] \leq \delta$),

$$\liminf_{\delta \to 0} \frac{\mathbb{E}_{\boldsymbol{\mu}}[\tau]}{\ln(1/\delta)} \geqslant \left(\sup_{\boldsymbol{\omega} \in \Sigma} \inf_{\boldsymbol{\lambda} \in \mathsf{Alt}(\boldsymbol{\mu})} \sum_{k=1}^{K} \frac{\omega_k (\mu_k - \lambda_k)^2}{2} \right)^{-1}. \tag{1}$$

- $\Sigma = \{\sum_{\boldsymbol{x} \in \mathcal{X}} w_{\boldsymbol{x}} \boldsymbol{x} : \boldsymbol{w} \in \Sigma_{|\mathcal{X}|}\}$: all possible arm allocations
- $\mathsf{Alt}(\boldsymbol{\mu}) = \{\boldsymbol{\lambda} \in \Lambda : \boldsymbol{\imath}^\star(\boldsymbol{\lambda}) \neq \boldsymbol{\imath}^\star(\boldsymbol{\mu})\}$: confusing parameters

(1)$\Rightarrow$ An optimal algorithm has a sampling strategy described by

$$\hat{\boldsymbol{\omega}}(t) \xrightarrow{t \to \infty} \boldsymbol{\omega}^\star(\boldsymbol{\mu}) := \arg\max_{\boldsymbol{\omega} \in \Sigma} F_{\boldsymbol{\mu}}(\boldsymbol{\omega}),$$

where $\hat{\boldsymbol{\omega}}(t)$ is the empirical allocation of arm draws up to round t, and

$$F_{\boldsymbol{\mu}}(\boldsymbol{\omega}) = \inf_{\boldsymbol{\lambda} \in \mathsf{Alt}(\boldsymbol{\mu})} \sum_{k=1}^{K} \frac{\omega_k (\mu_k - \lambda_k)^2}{2}.$$

A Computational Challenge for Matching Lower Bound

A standard way [Garivier and Kaufmann, 2016] consists of:

- (sampling rule) pull an action by tracking $\arg\max_{\boldsymbol{\omega}\in\Sigma} F_{\hat{\boldsymbol{\mu}}(t)}(\boldsymbol{\omega})$
- (stopping rule) Generalized Likelihood Ratio Test (GLRT):

$$\tau = \inf\left\{t : tF_{\hat{\boldsymbol{\mu}}(t)}(\hat{\boldsymbol{\omega}}(t)) > \ln\left(\frac{t}{\delta}\right) + o(1)\right\} \tag{2}$$

- (decision rule) return $\hat{\boldsymbol{i}} \leftarrow \boldsymbol{i}^\star(\hat{\boldsymbol{\mu}}(\tau))$

where $\hat{\boldsymbol{\mu}}(t)$ is the estimate of $\boldsymbol{\mu}$, and $\hat{\boldsymbol{\omega}}(t)$ is empirical pulled arm allocation up to t.

The value of $F_{\boldsymbol{\mu}}(\boldsymbol{\omega}) := \inf_{\boldsymbol{\lambda}\in\mathsf{Alt}(\boldsymbol{\mu})} \sum_{k=1}^{K} \frac{\omega_k(\mu_k-\lambda_k)^2}{2}$ is the most confusing parameter (`MCP`).

Difficulty in identifying MCP

Recall $\mathsf{Alt}(\boldsymbol{\mu}) = \{\boldsymbol{\lambda}\in\Lambda : \boldsymbol{i}^\star(\boldsymbol{\lambda}) \neq \boldsymbol{i}^\star(\boldsymbol{\mu})\}$

$\Rightarrow$ Prior approach is to solve $|\mathcal{X}|-1$ many convex programs by partitioning

$\mathsf{Alt}(\boldsymbol{\mu}) = \cup_{\boldsymbol{x}\neq\boldsymbol{i}^\star(\boldsymbol{\mu})}\{\boldsymbol{\lambda}\in\Lambda : \langle \boldsymbol{i}^\star(\boldsymbol{\mu}) - \boldsymbol{x}, \boldsymbol{\lambda}\rangle < 0\}$

However, $|\mathcal{X}| = \mathcal{O}(2^K)$!

Other Statistically Optimal Sampling Rules

- `FWS` [Wang et al., 2021]: apply the first-order methods to maximize $F_{\boldsymbol{\mu}}(\boldsymbol{\omega})$, where the gradient is derived by a `MCP` (Envelop theorem)
- `CombGame` [Jourdan et al., 2021]: need a `MCP` for chasing a saddle point.
- `ACC` [Qin and You, 2023]: need a `MCP` to get its best challenger.

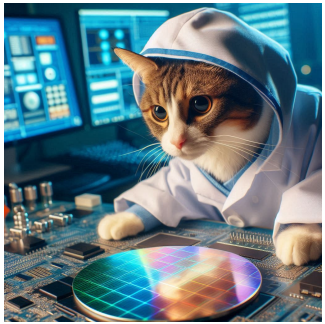


Figure 1: `MCP` in pure exploration resembles the role of the semi-conductor in modern technology.

Our Perturbed Frank-Wolfe Sampling (P-FWS)

- P-FWS uses perturbed Frank-Wolfe based algorithm to reach the optimal allocation
- All P-FWS needs are
  - (i) linear maximization (standard),
  - (ii) (ε, θ)-MCP, where $\varepsilon > 0, \theta \in (0, 1)$ (established in this work).

(ε, θ)-MCP

Return $\hat{F}$ s.t.

$$\mathbb{P}[F_{\boldsymbol{\mu}}(\boldsymbol{\omega}) \leq \hat{F} \leq (1+\varepsilon) F_{\boldsymbol{\mu}}(\boldsymbol{\omega})] \geq 1 - \theta$$

How to get (ε, θ)-MCP?

Let Us Observe MCP

For each suboptimal action $\boldsymbol{x} \neq i^\star(\boldsymbol{\mu})$, define

$$f_{\boldsymbol{x}}(\boldsymbol{\omega}, \boldsymbol{\mu}) := \inf_{\boldsymbol{\lambda} \in \mathbb{R}^K : \langle \boldsymbol{i}^\star(\boldsymbol{\mu}) - \boldsymbol{x}, \boldsymbol{\lambda} \rangle < 0} \sum_{k=1}^{K} \frac{\omega_k (\mu_k - \lambda_k)^2}{2}. \tag{3}$$

Property of $f_{\boldsymbol{x}}$ and its Lagrangian dual $g_{\boldsymbol{\omega},\boldsymbol{\mu}}$

$$f_{\boldsymbol{x}}(\boldsymbol{\omega}, \boldsymbol{\mu}) = \max_{\alpha \geqslant 0} g_{\boldsymbol{\omega},\boldsymbol{\mu}}(\boldsymbol{x}, \alpha) \text{ (strong duality)}$$

$g_{\boldsymbol{\omega},\boldsymbol{\mu}}(\boldsymbol{x}, \alpha)$ is *linear* in $\boldsymbol{x}$ and *concave* in α

$$\Rightarrow F_{\boldsymbol{\mu}}(\boldsymbol{\omega}) = \min_{\boldsymbol{x} \neq \boldsymbol{i}^\star(\boldsymbol{\mu})} f_{\boldsymbol{x}}(\boldsymbol{\omega}, \boldsymbol{\mu}) = \min_{\boldsymbol{x} \neq \boldsymbol{i}^\star(\boldsymbol{\mu})} \max_{\alpha \geqslant 0} g_{\boldsymbol{\omega},\boldsymbol{\mu}}(\boldsymbol{x}, \alpha) \tag{4}$$

(4) motivates us a design:

- $\boldsymbol{x}$-player employs a regret minimization algorithm to minimize $g_{\boldsymbol{\omega},\boldsymbol{\mu}}(\boldsymbol{x}, \alpha)$
- α-player employs a regret minimization algorithm to maximize $g_{\boldsymbol{\omega},\boldsymbol{\mu}}(\boldsymbol{x}, \alpha)$

Two-players Game

However, the choices are limited since...

1. $\{\boldsymbol{x} \neq \boldsymbol{i}^\star(\boldsymbol{\mu})\}$ is a discrete set which consist of $\mathcal{O}(2^K)$ elements
2. $\min\max = \max\min$ may not hold
3. *equilibrium* action $\boldsymbol{x}_e$ s.t. $F_{\boldsymbol{\mu}}(\boldsymbol{\omega}) = \max_{\alpha \geqslant 0} g_{\boldsymbol{\omega},\boldsymbol{\mu}}(\boldsymbol{x}_e, \alpha)$ need a maximizer in α
4. we want it to be computationally efficient, i.e. approximate $\boldsymbol{x}_e$ in polynomial time

Recall

$$F_{\boldsymbol{\mu}}(\boldsymbol{\omega}) = \min_{\boldsymbol{x} \neq \boldsymbol{i}^\star(\boldsymbol{\mu})} \max_{\alpha \geqslant 0} g_{\boldsymbol{\omega},\boldsymbol{\mu}}(\boldsymbol{x}, \alpha) = f_{\boldsymbol{x}_e}(\boldsymbol{\omega}, \boldsymbol{\mu})$$

Figure 2: Two-cats Game

Our (ε, θ)-MCP

Algorithm 1: (ε, θ)-MCP$(\boldsymbol{\omega}, \boldsymbol{\mu})$

for $n = 1, 2, \cdots$ **do**

(Follow-the-Perturbed-Leader) $\boldsymbol{\mathcal{Z}}_n \sim \exp(1)^K$ and $\eta_n = \frac{c_0}{\sqrt{n}}$

$$\boldsymbol{x}^{(n)} \in \arg\min_{\boldsymbol{x} \neq \boldsymbol{i}^\star(\boldsymbol{\mu})} \left(\sum_{m=1}^{n-1} g_{\boldsymbol{\omega},\boldsymbol{\mu}}(\boldsymbol{x}, \alpha^{(m)}) + \frac{\langle \boldsymbol{\mathcal{Z}}_n, \boldsymbol{x} \rangle}{\eta_n} \right)$$

(Best-Response) $\alpha^{(n)} \in \arg\max_{\alpha \geqslant 0} g_{\boldsymbol{\omega},\boldsymbol{\mu}}(\boldsymbol{x}^{(n)}, \alpha)$

if $\sqrt{n} > \dfrac{c_\theta(1+\varepsilon)}{\varepsilon \hat{F}}$, *where* $\hat{F} = \min_n g_{\boldsymbol{\omega},\boldsymbol{\mu}}(\boldsymbol{x}^{(n)}, \alpha^{(n)})$ **then return** $\hat{F}$;

end

Main Results

Theorem for (ε, θ)-MCP

Theorem 1

Let $(\boldsymbol{\omega}, \boldsymbol{\mu}) \in \Sigma_+ \times \Lambda$. The (ε, θ)-MCP$(\boldsymbol{\omega}, \boldsymbol{\mu})$ algorithm outputs $\hat{F}$:

- $\mathbb{P}\left[F_{\boldsymbol{\mu}}(\boldsymbol{\omega}) \leqslant \hat{F} \leqslant (1+\varepsilon)F_{\boldsymbol{\mu}}(\boldsymbol{\omega})\right] \geqslant 1 - \theta$
- the number of calls to linear maximization $(\boldsymbol{i}^\star(\cdot))$ is polynomial

P-FWS: the First poly (K)-time Optimal Algorithm

Theorem 2

Let $\boldsymbol{\mu} \in \Lambda$ and $\delta \in (0,1)$. P-FWS is δ-PAC, and

(i) $\limsup_{\delta \to 0} \frac{\mathbb{E}_{\boldsymbol{\mu}}[\tau]}{\ln(1/\delta)} \leq \left(\sup_{\boldsymbol{\omega} \in \Sigma} \inf_{\boldsymbol{\lambda} \in \mathsf{Alt}(\boldsymbol{\mu})} \sum_{k=1}^{K} \frac{\omega_k (\mu_k - \lambda_k)^2}{2} \right)^{-1}$

(ii) $\mathbb{E}_{\boldsymbol{\mu}}[\tau] = \text{poly}\,(K)$ in moderate confidence regime

(iii) the expected number of calls for linear maximization upper bounded by poly (K)

Note: To our best knowledge, (ii) is even not yet established in unstructured BAI.

Figure 3: A cat wins three medals in *Olympicat*.

Conclusion and Future Works

Conclusion: `P-FWS` closes the statistical-computational gap for combinatorial BAI by exploiting the structural properties in lower bound, and it show strong numerical performance.

Table 1: Averaged sample complexity at $\delta = 0.1$ over 100 independent runs on a graph with $|\mathcal{X}| = 343\,385$ spanning trees.

Algorithm	Sample Complexity
`P-FWS` (ours)	$1\,501$
`CombGame` [Jourdan et al., 2021] with our (ε, θ)-`MCP`	OOM

Future works: It remains largely unexplored whether one can close the computational-statistical gap for other distributions.

	reward distr.	comput. & stat. efficient	stat. optimal
comb. BAI semi-bandit	Gaussian	[Tzeng et al., 2023]	[Jourdan et al., 2021, Wang et al., 2021]
	Bernoulli	open	
comb. BAI bandit	Gaussian	open	[Wang et al., 2021]
	Bernoulli		
regret min.	Gaussian	open	[Combes et al., 2017]
	Bernoulli		

Table 2: Computational-statistical gap in combinatorial semi-bandits.

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